

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Operating Systems**

Code No: **CSC264** Semester: **4th Semester**

Full Marks: 60 Pass Marks: 24 Time: using SCAN, C-SCAN, Look and C-Look disk scheduling algorithm for processing the following requests queue: 35, 70, 45, 15, 65, 20, 80, 90, 75, 130. Suppose the disk has tracks numbered from 0 to 150 and assume the disk arm to be at 30 and moving outward. 3. Explain Sleeping Barber problem. Illustrate on how it can be solved. Section B Attempt any Eight questions [8×5=40] 4. Explain microkernels and exokernels. 5. Consider a swapping system in which memory consists of the following hole sizes in memory order 15 MB, 2 MB, 10 MB, 6 MB, 8 MB and 20 MB. Which hole is taken for successive segment requests of a) 6 MB b) 10 MB c) 8 MB for first fit, next fit and best fit. 6. Explain how semaphore solves the problem of critical section? 7. How do you think deadlock can be avoided? Explain. 8. Explain Inter-Process Communication in Linux. 9. List different file structures and explain them. 10. Calculate the average waiting time and turn around time using priority algorithm (priority 1 being the highest) for the given scenario: PID Burst Time (s) Arrival Time Priority
A 40 3 B 22 3 C 42 2 D 23 1 11. Explain memory mapped IO. 12. Write short notes on a) Virtual memory b) Race condition

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain the translation of logical address into physical address using segment table with necessary diagram. List advantages and disadvantages of segmentation.
2. Find the seek time using SCAN, C-SCAN, Look and C-Look disk scheduling algorithm for processing the following requests queue: 35, 70, 45, 15, 65, 20, 80, 90, 75, 130. Suppose the disk has tracks numbered from 0 to 150 and assume the disk arm to be at 30 and moving outward.
3. Explain Sleeping Barber problem. Illustrate on how it can be solved.

Section B

Attempt any Eight questions [8×5=40]

4. Explain microkernels and exokernels.
5. Consider a swapping system in which memory consists of the following hole sizes in memory order 15 MB, 2 MB, 10 MB, 6 MB, 8 MB and 20 MB. Which hole is taken for successive segment requests of a) 6 MB b) 10 MB c) 8 MB for first fit, next fit and best fit.
6. Explain how semaphore solves the problem of critical section?
7. How do you think deadlock can be avoided? Explain.
8. Explain Inter-Process Communication in Linux.
9. List different file structures and explain them.

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10. Calculate the average waiting time and turn around time using priority algorithm (priority 1 being the highest) for the given scenario:

PID	Burst Time (s)	Arrival Time	Priority
A	4	0	3
B	2	2	2
C	4	2	2
D	2	3	1
 11. Explain memory mapped IO.
 12. Write short notes on a) Virtual memory b) Race condition